

NET 511: Week 1 Practical Guide: Exploring the OSI Model & TCP/IP (5 Marks).

1. Map Protocols to OSI/TCP/IP Layers

Create a small table mapping these protocols:

- Ethernet - Layer 2 Data link
- ARP - Layer 2 Data link
- IP – Layer 3 Network
- ICMP - Layer 3 Network
- TCP/UDP - Layer 4 Transport
- DNS, HTTP/HTTPS - Layer 7 Application

Submit: Screenshot or table in document.

2. Packet Encapsulation in Wireshark

1. Start Wireshark capture
2. Browse any website
3. Stop capture
4. Click any packet → expand:
 - Frame
 - Ethernet II
 - IP
 - TCP/UDP
 - Application (if present)

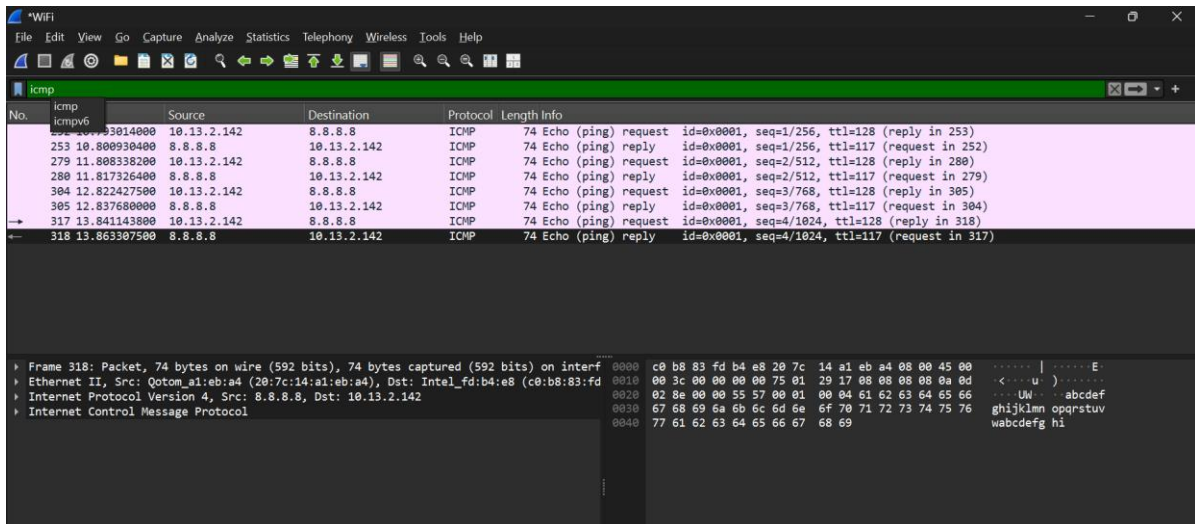
Submit: Screenshot showing all expanded layers + 2–3 sentences explaining encapsulation.

3. Observe Specific Protocols in Wireshark ICMP

Run ping 8.8.8.8

Wireshark filter: icmp

Submit: Screenshot of Echo Request & Reply.



ARP

Run arp -a or ping a device

Wireshark filter: arp

Submit: Screenshot of ARP Request & Reply.

DNS

Run nslookup microsoft.com

Wireshark filter: dns

Submit: Screenshot of DNS Query & Response.

TCP Handshake

Visit any website

Wireshark filter: tcp

Find packets labeled SYN → SYN-ACK → ACK

Submit: Screenshot of the 3-way handshake.

WEEK 2: Network Topologies

Star Topology

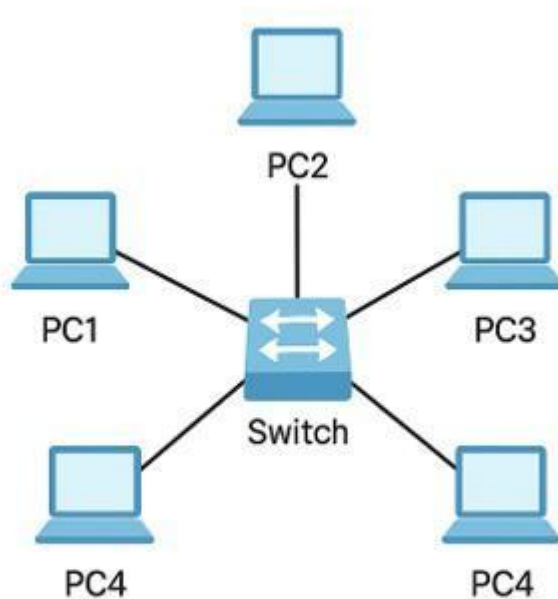
Topology Overview

The Star topology consists of multiple end devices connected to a central switch. Each PC communicates through the switch, which acts as the core connection point. **Topology**

Characteristics

- Centralized communication point
- Easy to manage and expand
- Failure of individual links affects only the connected device
- Switch failure impacts the entire topology

Topology Diagram



Star Topology

IP Addressing for Star Topology

Example network: 192.168.1.0/24

<u>Device IP Address</u>	
<u>PC1</u>	<u>192.168.1.10</u>
<u>PC2</u>	<u>192.168.1.11</u>
<u>PC3</u>	<u>192.168.1.12</u>
<u>PC4</u>	<u>192.168.1.13</u> <u>Expected</u>

result:

All devices should communicate successfully through the switch.

Hint: perform ping tests

Partial Mesh Topology

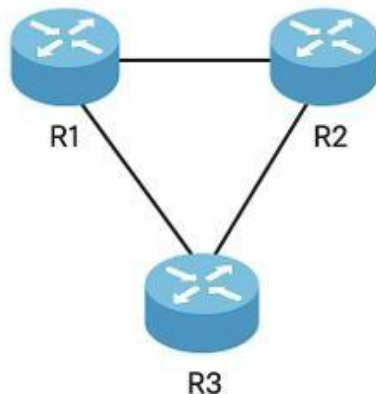
Topology Overview

A Partial Mesh topology connects routers so that some—but not all—devices have multiple redundant paths. This improves fault tolerance while reducing cabling and configuration complexity.

Topology Characteristics

- More resilient than star topology
- Provides alternate routes if a link fails
- Lower cost than full mesh
- Suitable for WAN or inter-router networking

Diagram (Partial Mesh Topology)



Partial Mesh Topology

IP Address Assignment for Router Links

Each router-to-router link uses a **/30 network**, providing exactly two usable addresses.

Link	Network	Router A IP	Router B IP
R1–R2	10.1.1.0/30	10.1.1.1	10.1.1.2
R1–R3	10.1.1.4/30	10.1.1.5	10.1.1.6
R2–R3	10.1.1.8/30	10.1.1.9	10.1.1.10

Example Cisco Interface Configuration

R1 interface s0/0/0 ip address 10.1.1.1

255.255.255.252 no shutdown

interface s0/0/1 ip address 10.1.1.5

255.255.255.252 no shutdown

Connectivity Tests

From each router:

- ping 10.1.1.2
- ping 10.1.1.6
- ping 10.1.1.10

write points on how to do Redundancy and Fault Tolerance Assessment

1. Identify Critical Network Devices

Start by listing which devices are essential for network operation:

- Core routers (routing between networks)
- Switches (especially central ones in star topology)
- Links between routers in partial mesh
- Servers (DHCP, DNS, HTTP if used)

This helps you know what must be tested for failure scenarios.

2. Document the Normal Network Path

Before testing anything:

- Run **ping tests between all networks**
- Use **tracert/traceroute** to map the normal path
- Record which routes are primary

This gives a baseline to compare against after failures.

3. Test Single Point of Failure (Star Topology)

In a star topology:

- Disable the central switch or router
- Observe if all connected devices lose connectivity
- Note: Star networks usually fail completely if the center device fails

Write down:

- What stopped working
- Which devices became isolated

4. Test Link Failures (Partial Mesh)

In partial mesh:

- Shut down one router-to-router link at a time
- Check if traffic reroutes through alternate paths
- Use continuous ping to monitor packet loss

Key idea:

- Partial mesh should still function even if one link fails

5. Verify Routing Protocol Behaviour

If dynamic routing is used (like RIP, OSPF):

- Check if routes automatically update
- Use show ip route before and after failure
- Confirm convergence time (how fast network recovers)

6. Check Redundant Paths

- Identify backup routes in topology
- Force primary path failure
- Confirm traffic switches to secondary route

This is the core of redundancy testing.

7. Measure Fault Tolerance

Evaluate:

- How many failures the network can handle before breakdown
- Whether communication is fully or partially maintained
- Impact on latency or packet loss during rerouting

8. Use Simulation Tools in Packet Tracer

- Use **Simulation Mode**
- Watch packets move through network during failure
- Observe dropped vs rerouted packets

9. Record Results Clearly

For each test, document:

- What failed (device/link)
- Expected outcome
- Actual outcome
- Recovery behaviour

10. Conclude Network Reliability

Summarize:

- Star topology: low redundancy, high dependency on central device
- Partial mesh: better redundancy, multiple paths improve fault tolerance

Student Lab Handout: Network Protocols and Services

Total Marks: 10

Lab Overview

In this lab, you will configure and test common network services used in enterprise networks.

You must follow each step carefully and provide screenshots as evidence for assessment.

Lab Environment

You may use Cisco Packet Tracer, GNS3, or an approved lab environment.

Required devices:

- One router
- One server
- One client PC

Part 1: DHCP Configuration (2 Marks)

Step 1: Log into the router or server that will provide DHCP services.

Step 2: Enable the DHCP service.

Step 3: Configure the following DHCP options:

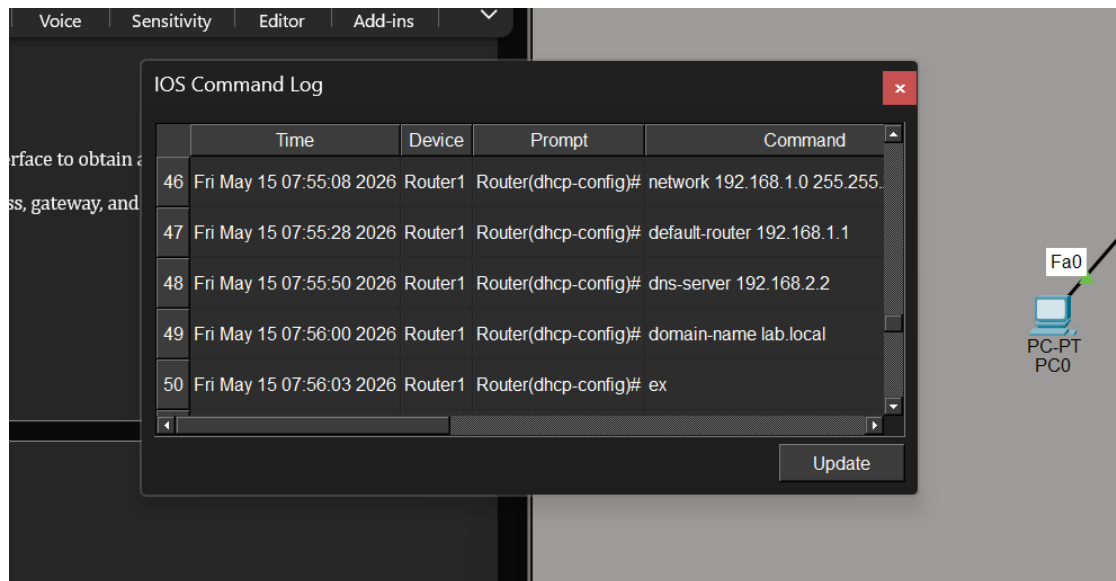
- Network address
- Subnet mask
- Default gateway
- DNS server address

Step 4: On the client PC, set the network interface to obtain an IP address automatically.

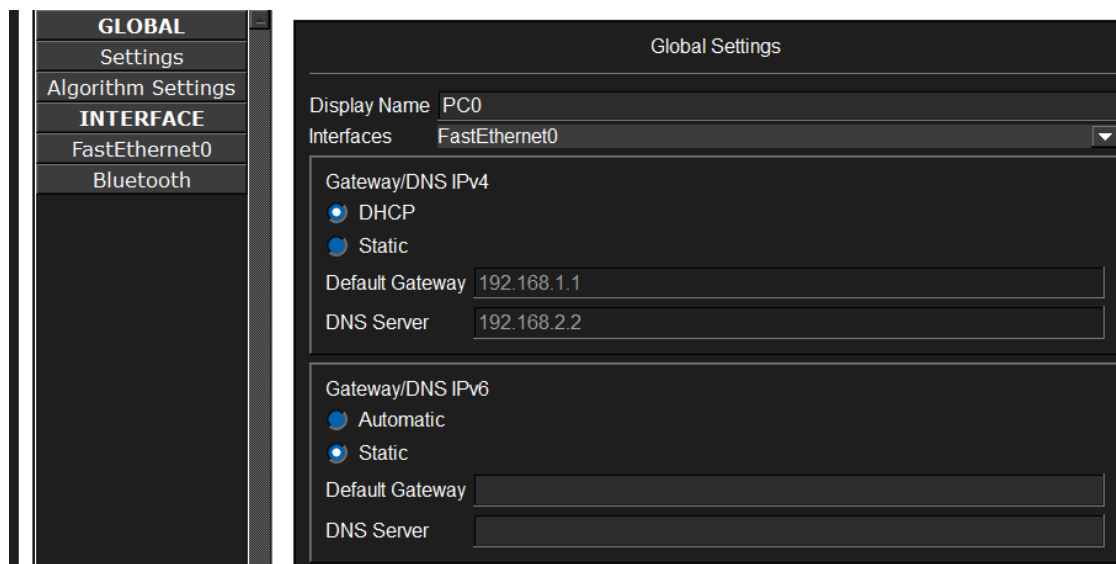
Step 5: Verify the client receives an IP address, gateway, and DNS server.

Evidence Required:

- Screenshot of DHCP configuration



- Screenshot showing the client IP address



Part 2: DNS Configuration (2 Marks)

Step 1: Open the DNS service on the server.

Step 2: Create a DNS A record (e.g., www.lab.local).

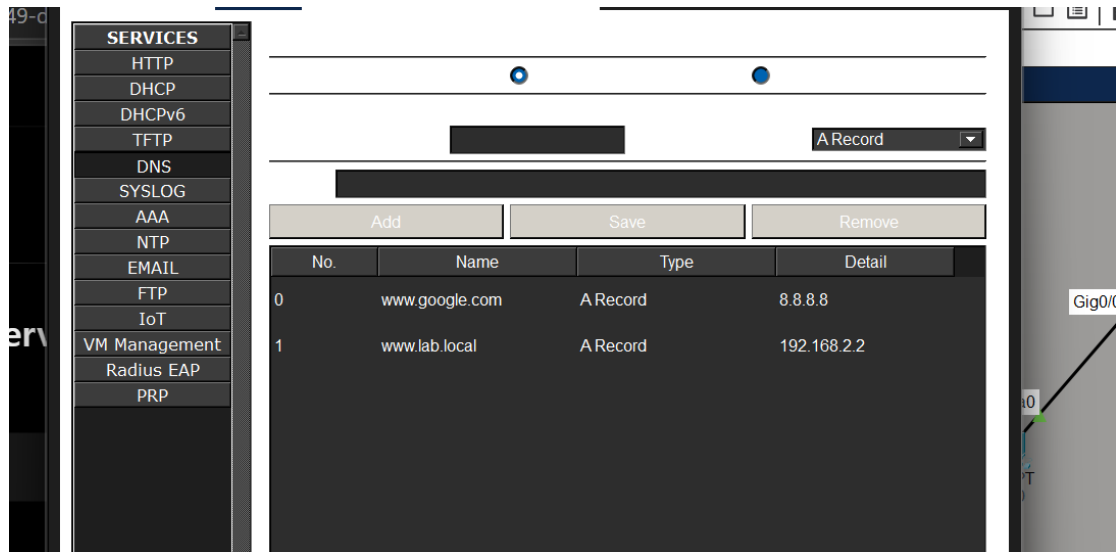
Step 3: Ensure the client is configured to use the DNS server.

Step 4: On the client, open the command prompt. Step

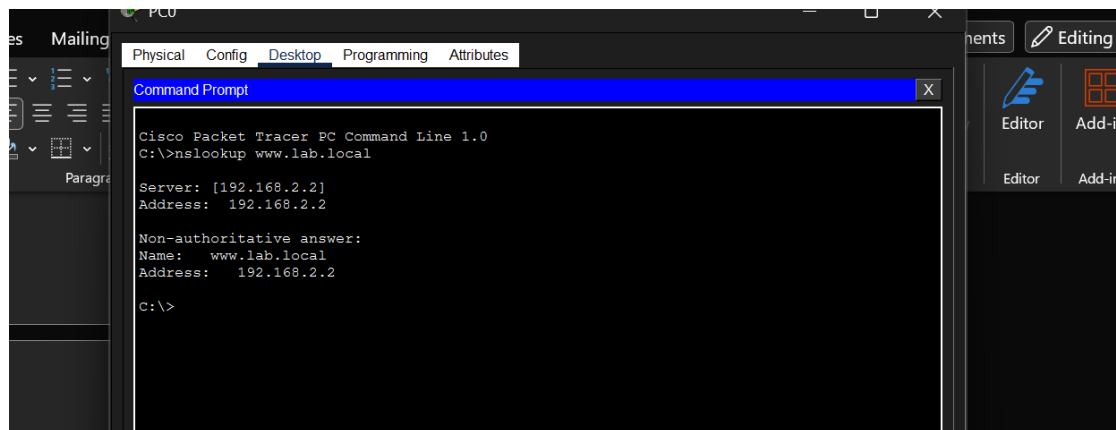
5: Run the command: nslookup www.lab.local

Evidence Required:

- Screenshot of DNS configuration -



Screenshot of successful nslookup output



Part 3: NTP Configuration (2 Marks)

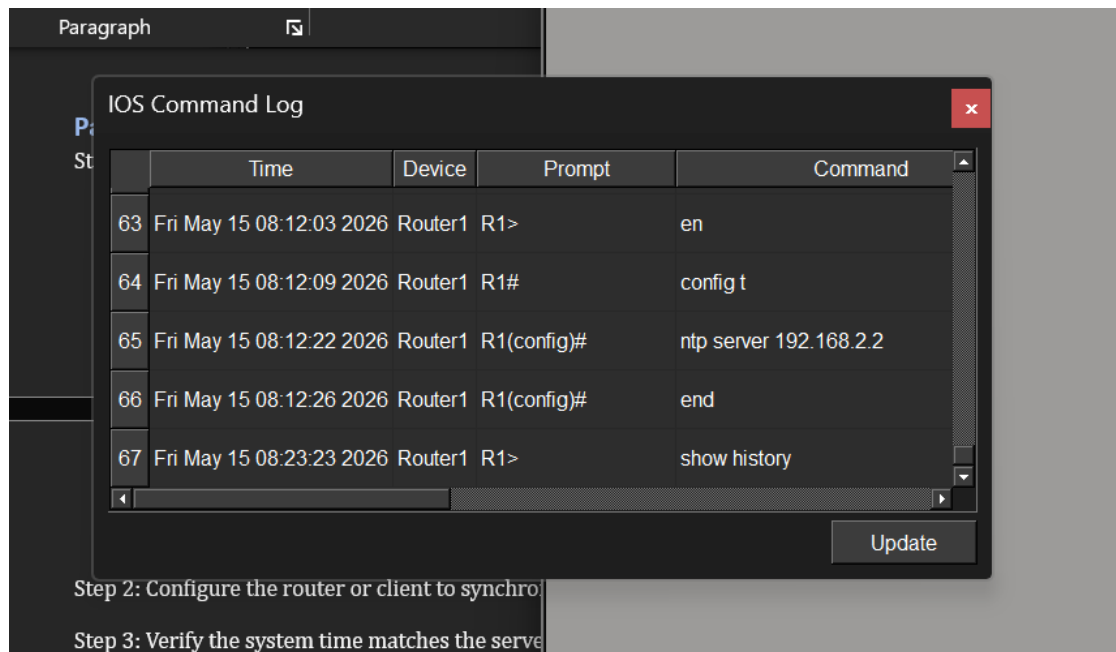
Step 1: Enable NTP on the server.

Step 2: Configure the router or client to synchronise with the NTP server.

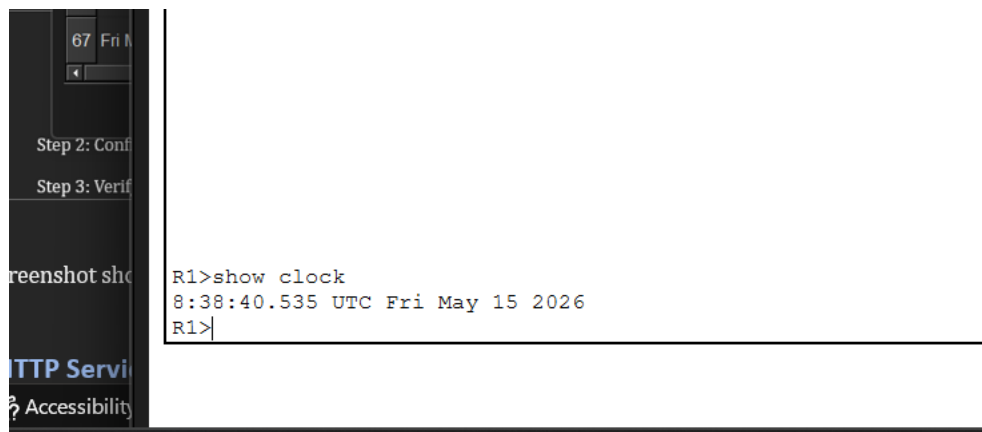
Step 3: Verify the system time matches the server time.

Evidence Required:

- Screenshot of NTP configuration –



Screenshot showing correct system time



Part 4: HTTP Service Configuration (2 Marks)

Step 1: Enable the HTTP service on the server.

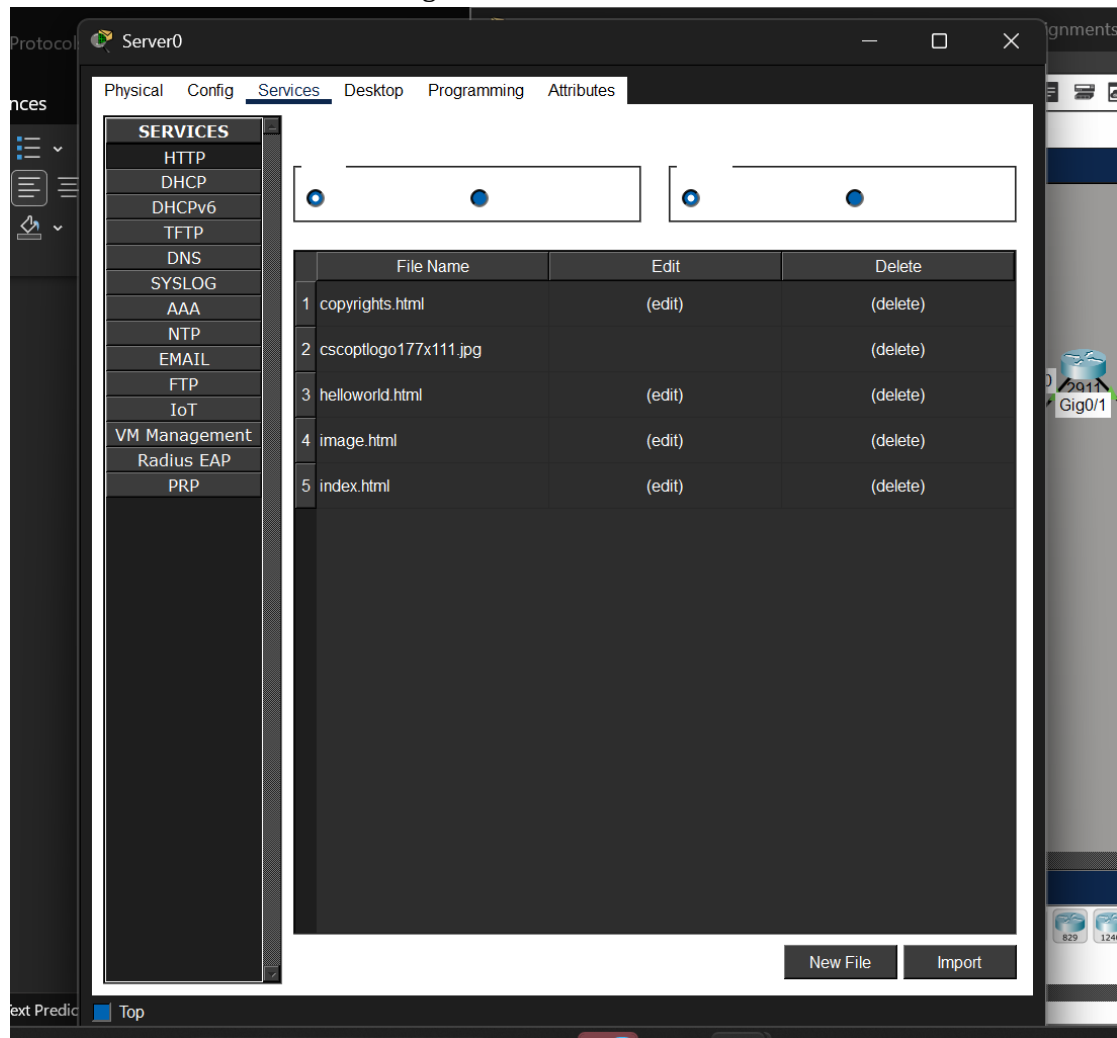
Step 2: Verify that a default web page is available.

Step 3: From the client, open a web browser.

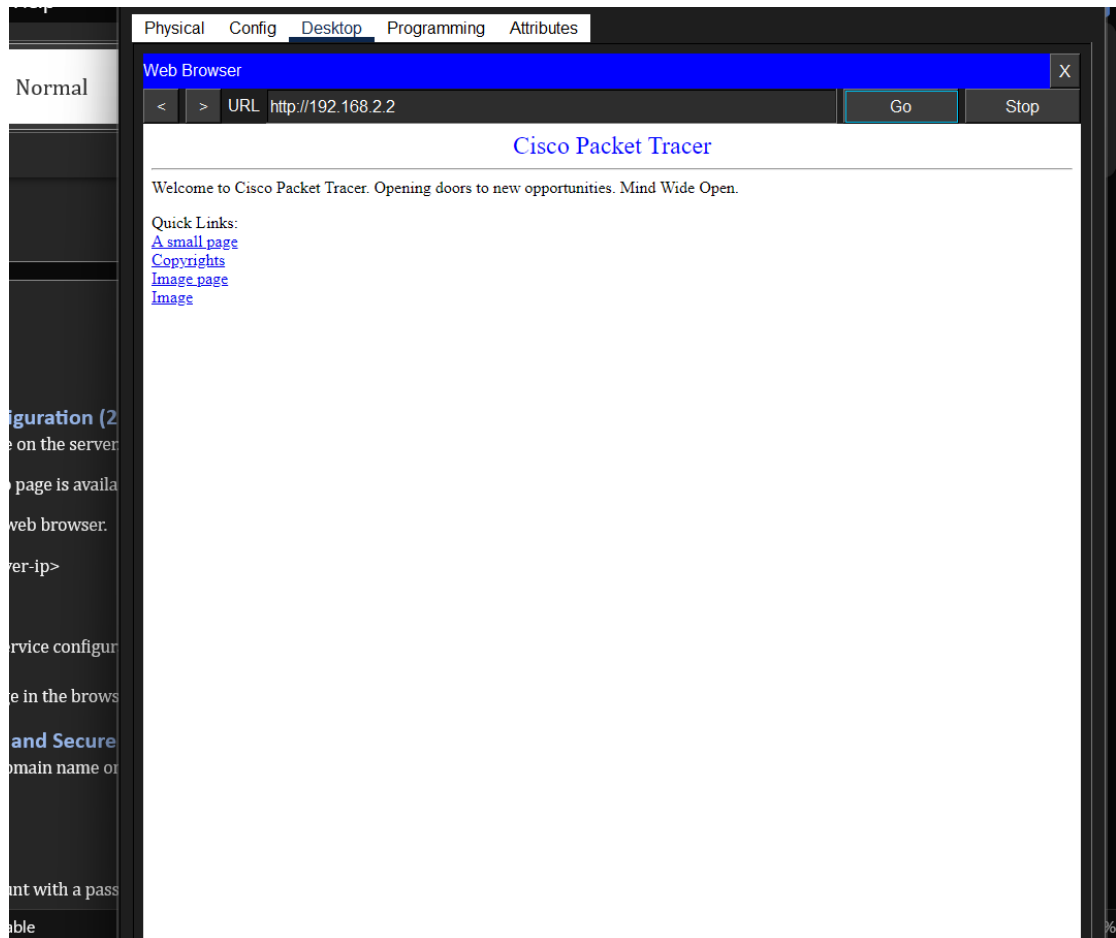
Step 4: Navigate to `http://<server-ip>`

Evidence Required:

- Screenshot of HTTP service configuration



- Screenshot of web page in the browser



Part 5: SSH Configuration and Secure Access (2 Marks)

Step 1: Set the hostname and domain name on the router or server.

Step 2: Generate RSA keys.

Step 3: Enable SSH version 2.

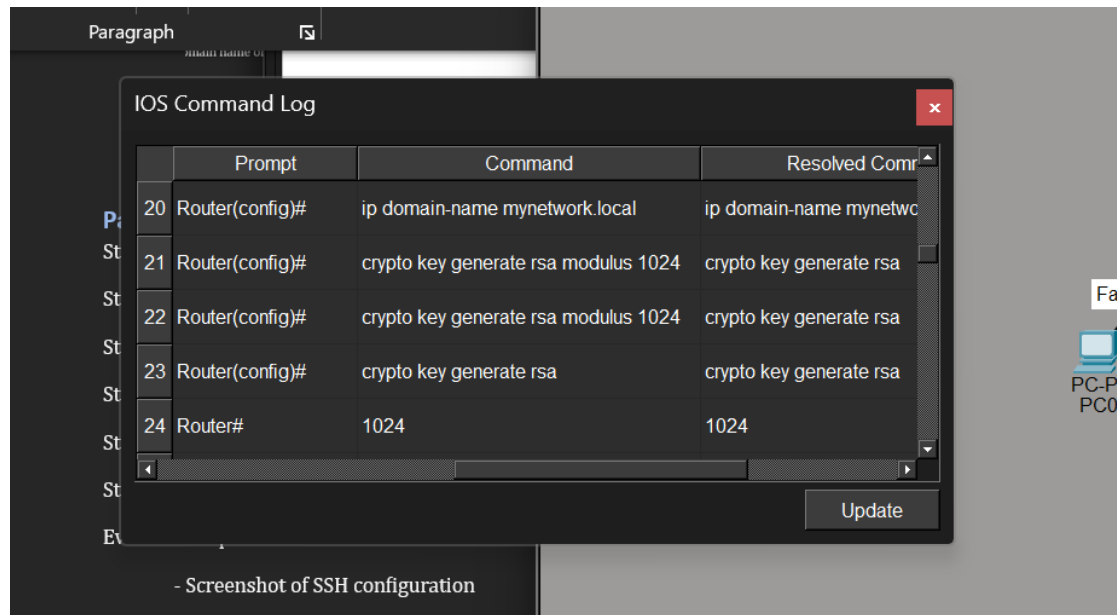
Step 4: Create a local user account with a password.

Step 5: Disable Telnet access.

Step 6: From the client, connect using: `ssh username@device-ip`

Evidence Required:

- Screenshot of SSH configuration



- Screenshot of successful SSH login

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>nslookup www.lab.local

Server: [192.168.2.2]
Address: 192.168.2.2

Non-authoritative answer:
Name: www.lab.local
Address: 192.168.2.2

C:\>ssh -l Cisco123 192.168.2.2

% Connection refused by remote host
C:\>ssh -l admin 192.168.2.2

% Connection refused by remote host
C:\>ssh -l $1$mERr$RldxcCZEZsTFtETUyRaA50 192.168.2.2

% Connection refused by remote host
C:\>ssh -l 5 $1$mERr$RldxcCZEZsTFtETUyRaA50 192.168.2.2
Invalid Command.

C:\>
```

Submission Checklist

Ensure all screenshots are clear, readable, and labelled.

Missing evidence will result in lost marks.

Applied IPv4 Subnetting – Practical Demonstration

Student Name: _____

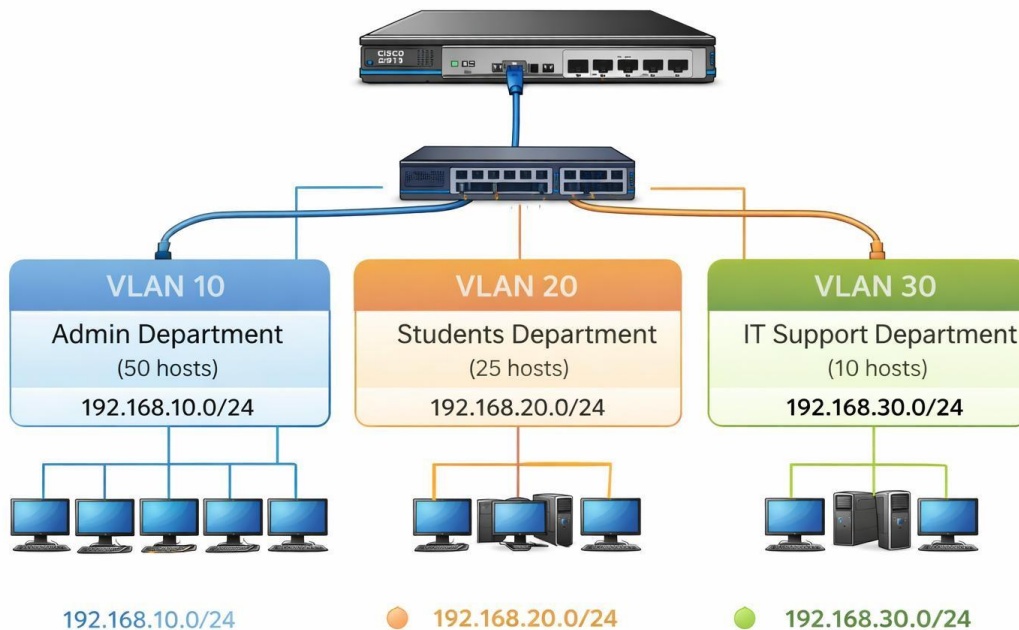
Course / Module: _____

Date: _____

1. Objective

The objective of this practical is to demonstrate applied IPv4 subnetting using VLSM, configure inter-VLAN routing using Router-on-a-Stick, assign IP addresses to end devices, and verify connectivity between VLANs.

2. Network Scenario



The network consists of one router, one switch, and three VLANs as shown below:

- VLAN 10 – Admin Department (50 hosts)
- VLAN 20 – Students Department (25 hosts)
- VLAN 30 – IT Support Department (10 hosts)

Base Network Address: 192.168.10.0/24

3. VLSM Subnet Plan

The subnetting was performed using Variable Length Subnet Masking (VLSM).

VLAN 10 (Admin)

Network Address: 192.168.10.0/26

Usable Range: 192.168.10.1 – 192.168.10.62

Default Gateway: 192.168.10.1

VLAN 20 (Students)

Network Address: 192.168.10.64/27

Usable Range: 192.168.10.65 – 192.168.10.94

Default Gateway: 192.168.10.65

VLAN 30 (IT Support)

Network Address: 192.168.10.96/28

Usable Range: 192.168.10.97 – 192.168.10.110

Default Gateway: 192.168.10.97

Network ID	Subnet mask	Available hosts	VLAN	Default gateway	PC
192.168.10.0/26	255.255.255.192	64	10 (Admin)	192.168.10.1	192.168.10.10
192.168.10.64/27	255.255.255.224	32	20 (Student)	192.168.10.65	192.168.10.70
192.168.10.96/28	255.255.255.240	16	30 (IT support)	192.168.10.97	192.168.10.100

4. Router Configuration (Router-on-a-Stick)

The router subinterfaces were configured as follows:

```
interface g0/0.10
```

```
encapsulation dot1Q 10
```

```
ip address 192.168.10.1 255.255.255.192
```

```
interface g0/0.20
```

```
encapsulation dot1Q 20
```

```
ip address 192.168.10.65 255.255.255.224
```

```
interface g0/0.30
```

```
encapsulation dot1Q 30
```

```
ip address 192.168.10.97 255.255.255.240
```

5. Switch Configuration

VLANs were created and assigned to switch ports. The router-facing port was configured as a trunk.

6. PC IP Address Assignment

Admin PC: 192.168.10.10 /26, Gateway 192.168.10.1

Student PC: 192.168.10.70 /27, Gateway 192.168.10.65

IT Support PC: 192.168.10.100 /28, Gateway 192.168.10.97

7. Connectivity Testing

Ping tests were performed between VLANs to verify inter-VLAN routing. All pings were successful.

8. Conclusion

This practical successfully demonstrated VLSM subnetting, Router-on-a-Stick configuration, and inter-VLAN connectivity. All configuration objectives were met and verified through testing.

IPv4 and IPv6 Addressing (Dual-Stack)

Objectives

- Configure dual-stack (IPv4 and IPv6) addressing on routers and hosts
- Validate connectivity and default gateways
- Inspect ARP vs ICMPv6 Neighbor Discovery
- Submit evidence using ping tests, packet captures, and show commands

1. Dual-Stack Overview

Dual-stack networking allows IPv4 and IPv6 to operate simultaneously on the same network devices. This approach ensures backward compatibility while enabling a smooth transition from IPv4 to IPv6.

2. Router Dual-Stack Configuration

Example Network:

IPv4 Network: 192.168.1.0/24

IPv6 Network: 2001:db8:1::/64 **Cisco**

Router Configuration:

```
interface GigabitEthernet0/0 ip  
address 192.168.1.1 255.255.255.0  
ipv6 address 2001:db8:1::1/64  
ipv6 enable  
no shutdown
```

3. Host Dual-Stack Configuration

- ✦ IPv4 Address: 192.168.1.10/24
- ✦ IPv4 Gateway: 192.168.1.1
- ✦ IPv6 Address: 2001:db8:1::10/64
- ✦ IPv6 Gateway: 2001:db8:1::1 (learned via Router Advertisement)

4. Connectivity and Gateway Validation

IPv4 Ping Tests:

- **ping 192.168.1.1**
- **ping 192.168.1.10**

IPv6 Ping Tests:

- **ping 2001:db8:1::1**
- **ping 2001:db8:1::10**

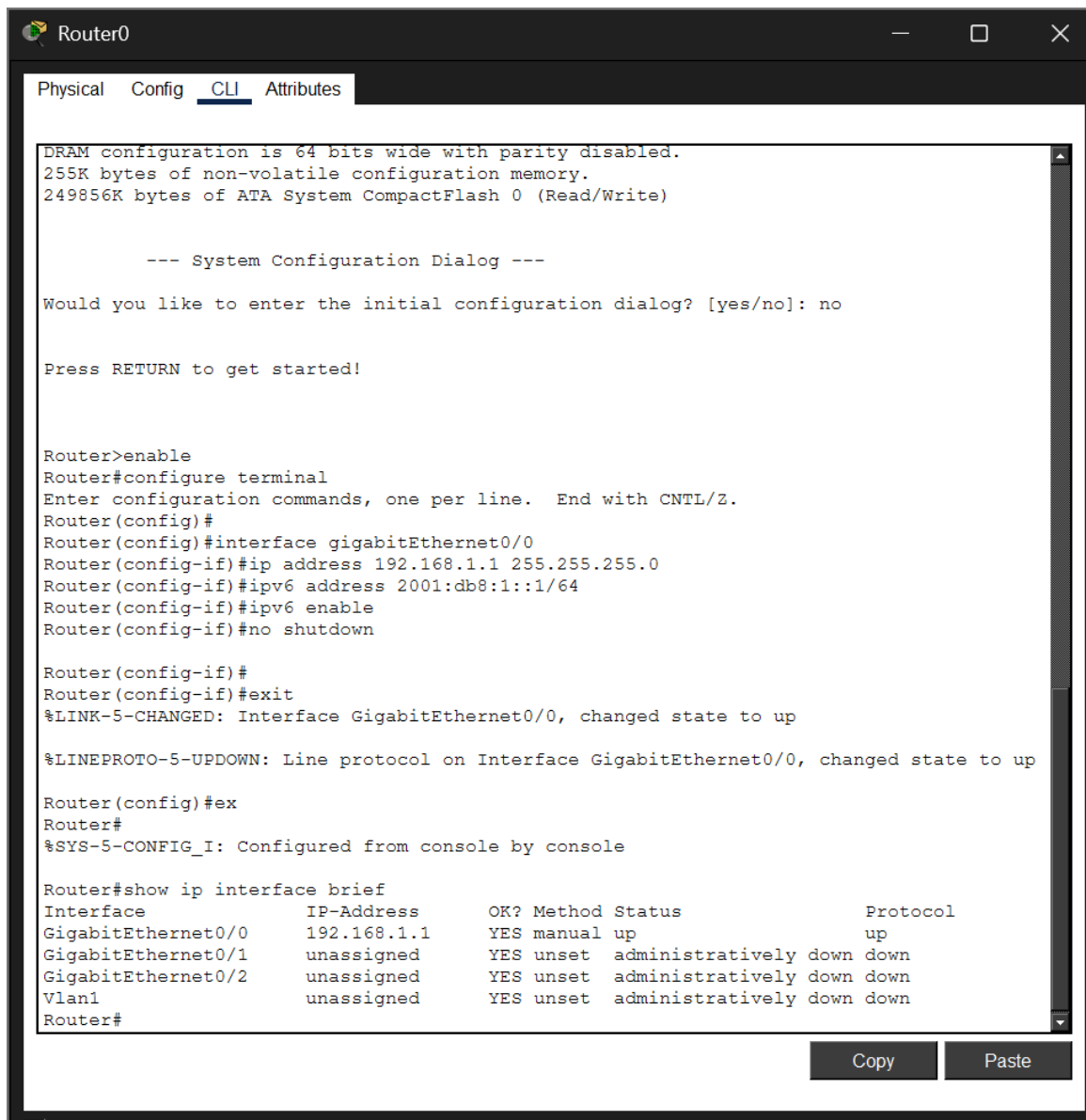
5. ARP vs ICMPv6 Neighbor Discovery

ARP is used in IPv4 networks to map IP addresses to MAC addresses using broadcast messages. IPv6 replaces ARP with ICMPv6 Neighbor Discovery, which uses multicast messages and provides better security and efficiency.

6. Evidence and Verification Commands

Router Commands:

- **show ip interface
brief**



The screenshot shows a Cisco Router CLI window titled "Router0". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The terminal output shows the following sequence of commands and responses:

```
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ipv6 address 2001:db8:1::1/64
Router(config-if)#ipv6 enable
Router(config-if)#no shutdown

Router(config-if)#
Router(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.1.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

Router#

At the bottom right of the window, there are "Copy" and "Paste" buttons.

- **show ipv6
interface brief**

Router0

Physical Config CLI Attributes

```
Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#interface gigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ipv6 address 2001:db8:1::1/64
Router(config-if)#ipv6 enable
Router(config-if)#no shutdown

Router(config-if)#
Router(config-if)#exit
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0       192.168.1.1     YES manual up              up
GigabitEthernet0/1       unassigned      YES unset  administratively down down
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down

Router#show ipv6 interface brief
GigabitEthernet0/0       [up/up]
FE80::20B:BEFF:FE69:1001
2001:DB8:1::1
GigabitEthernet0/1       [administratively down/down]
unassigned
GigabitEthernet0/2       [administratively down/down]
unassigned
Vlan1                    [administratively down/down]
unassigned
Router#
```

Copy Paste

- **show ipv6**

neighbors

```

unassigned
Vlan1                                [administratively down/down]
unassigned
Router#ping 2001:db8:1::10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:db8:1::10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#show run | include ipv6
ipv6 unicast-routing
no ipv6 cef
  ipv6 address 2001:DB8:1::1/64
  ipv6 enable
Router#show ip interface brief
Interface      IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0  192.168.1.1    YES manual up              up
GigabitEthernet0/1  unassigned     YES unset  administratively down down
GigabitEthernet0/2  unassigned     YES unset  administratively down down
GigabitEthernet0/0/0 unassigned     YES unset  administratively down down
Vlan1           unassigned     YES unset  administratively down down
Router#ping 2001:db8:1::1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:db8:1::1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/7/23 ms

Router#show ipv6 neighbors
Router#ping 2001:db8:1::10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:db8:1::10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#show ipv6 neighbors
IPv6 Address                               Age Link-layer Addr State Interface
2001:DB8:1::10                             1 0060.5CD6.350B REACH Gig0/0
Router#

```

Copy

Paste

Host Commands:

- **ipconfig /all**
(Windows)

```

Reply from 192.168.1.10: bytes=32 time=9ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=7ms TTL=128
Reply from 192.168.1.10: bytes=32 time=16ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 16ms, Average = 8ms

C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address. . . . .: 000D.BD88.0C19
    Link-local IPv6 Address . . . . .: FE80::20D:BDFF:FE88:C19
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 192.168.1.10
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.1.1
    DHCP Servers . . . . .: 0.0.0.0
    DHCPv6 IAID . . . . .:
    DHCPv6 Client DUID. . . . .: 00-01-00-01-80-3A-02-C0-00-0D-BD-88-0C-19
    DNS Servers . . . . .: ::
                                   0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Physical Address. . . . .: 0030.F2CC.B181
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0
    DHCP Servers . . . . .: 0.0.0.0
    DHCPv6 IAID . . . . .:
    DHCPv6 Client DUID. . . . .: 00-01-00-01-80-3A-02-C0-00-0D-BD-88-0C-19
    DNS Servers . . . . .: ::
                                   0.0.0.0

```

Top

Conclusion

The successful configuration and validation of IPv4 and IPv6 dual-stack networking ensures reliable connectivity, proper gateway functionality, and an understanding of address resolution mechanisms in both protocols.

Week 7 Lab: Data Center Architecture & Cloud Concepts

Objective

Design and implement a multi-tier data center network with VLAN segmentation, interVLAN routing, and gateway redundancy using HSRP/VRRP. Validate connectivity and provide evidence.

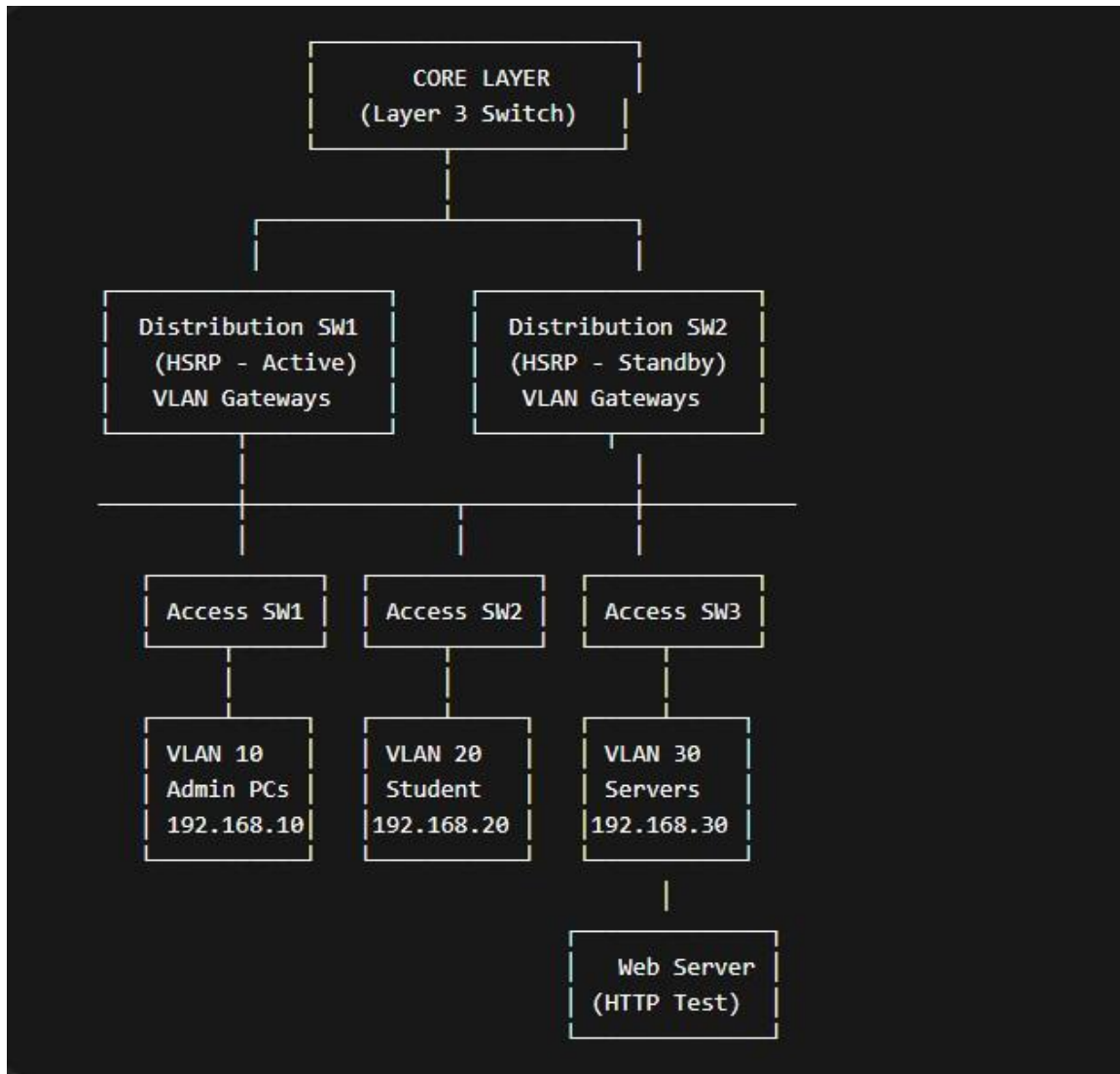
Topology Overview

Three-tier architecture:

- Core Layer – High-speed backbone
- Distribution Layer – Routing and redundancy (HSRP/VRRP)
- Access Layer – End devices (PCs/Servers)

VLANs:

- VLAN 10 – Admin (192.168.10.0/24)
- VLAN 20 – Students (192.168.20.0/24)
- VLAN 30 – Servers (192.168.30.0/24)



IP Addressing Table

VLAN	Network	Gateway (Virtual IP)
VLAN 10	192.168.10.0/24	192.168.10.1
VLAN 20	192.168.20.0/24	192.168.20.1
VLAN 30	192.168.30.0/24	192.168.30.1

Implementation Tasks

1. 1. Configure VLANs and Access Ports

Create VLANs 10, 20, 30 and assign switch ports accordingly.

2. 2. Configure Inter-VLAN Routing (SVIs)

Create VLAN interfaces on Layer 3 switches.

3. 3. Configure HSRP/VRRP Redundancy

Set virtual IPs and priorities for active/standby roles.

4. 4. Validate Connectivity

Test VLAN communication, inter-VLAN routing, and failover.

CLI Configuration Examples

VLAN Configuration

```
vlan 10 name  
Admin vlan 20  
name Students  
vlan 30 name  
Servers
```

Access Port Assignment

```
interface fa0/1  
switchport mode access  
switchport access vlan 10
```

```
interface fa0/2  
switchport mode access  
switchport access vlan 20
```

```
interface fa0/3  
switchport mode access  
switchport access vlan 30
```

```
SVI Configuration interface vlan 10 ip  
address 192.168.10.2 255.255.255.0 no  
shutdown
```

```
interface vlan 20 ip address  
192.168.20.2 255.255.255.0 no  
shutdown
```

```
interface vlan 30 ip address  
192.168.30.2 255.255.255.0 no  
shutdown
```

HSRP Configuration (Switch 1 - Active)

```
interface vlan 10 standby 10 ip  
192.168.10.1 standby 10 priority 110  
standby 10 preempt
```

```
interface vlan 20 standby  
20 ip 192.168.20.1 standby  
20 priority 110 standby 20  
preempt
```

```
interface vlan 30 standby  
30 ip 192.168.30.1 standby  
30 priority 110 standby 30  
preempt
```

HSRP Configuration (Switch 2 - Standby)

```
interface vlan 10 standby  
10 ip 192.168.10.1 standby  
10 priority 100 standby 10  
preempt
```

```
interface vlan 20 standby  
20 ip 192.168.20.1 standby  
20 priority 100 standby 20  
preempt
```

```
interface vlan 30 standby  
30 ip 192.168.30.1
```

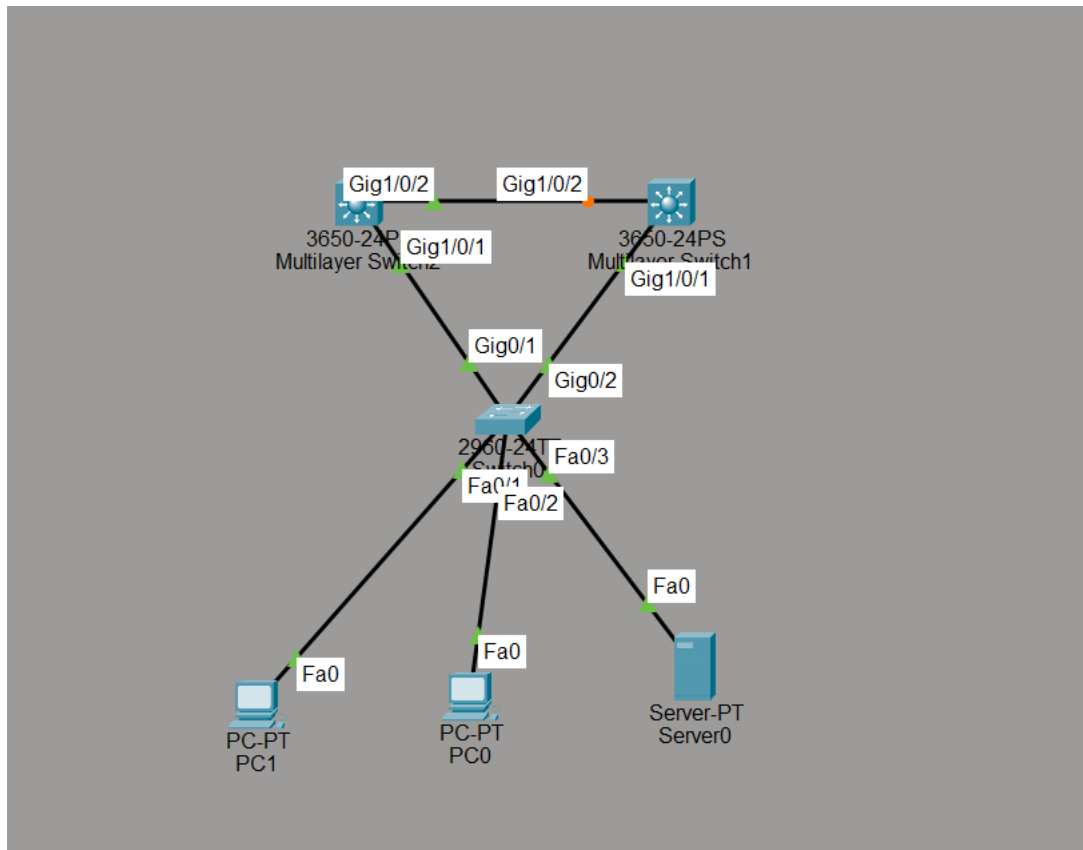
```
standby 30 priority 100  
standby 30 preempt
```

Testing & Validation

- Ping between VLANs
- Access web server via HTTP
- Simulate failure (shutdown active switch)
- Verify failover using: show standby brief

Submission Requirements

- Network topology diagram



- Configuration screenshots
Show ip interface brief

Switch0

PhysicalConfigCLIAttributes

%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/18, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/22, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/23, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to administratively down
SW-Access(config-if-range)#end
SW-Access#write memory
%SYS-5-CONFIG_I: Configured from console by console

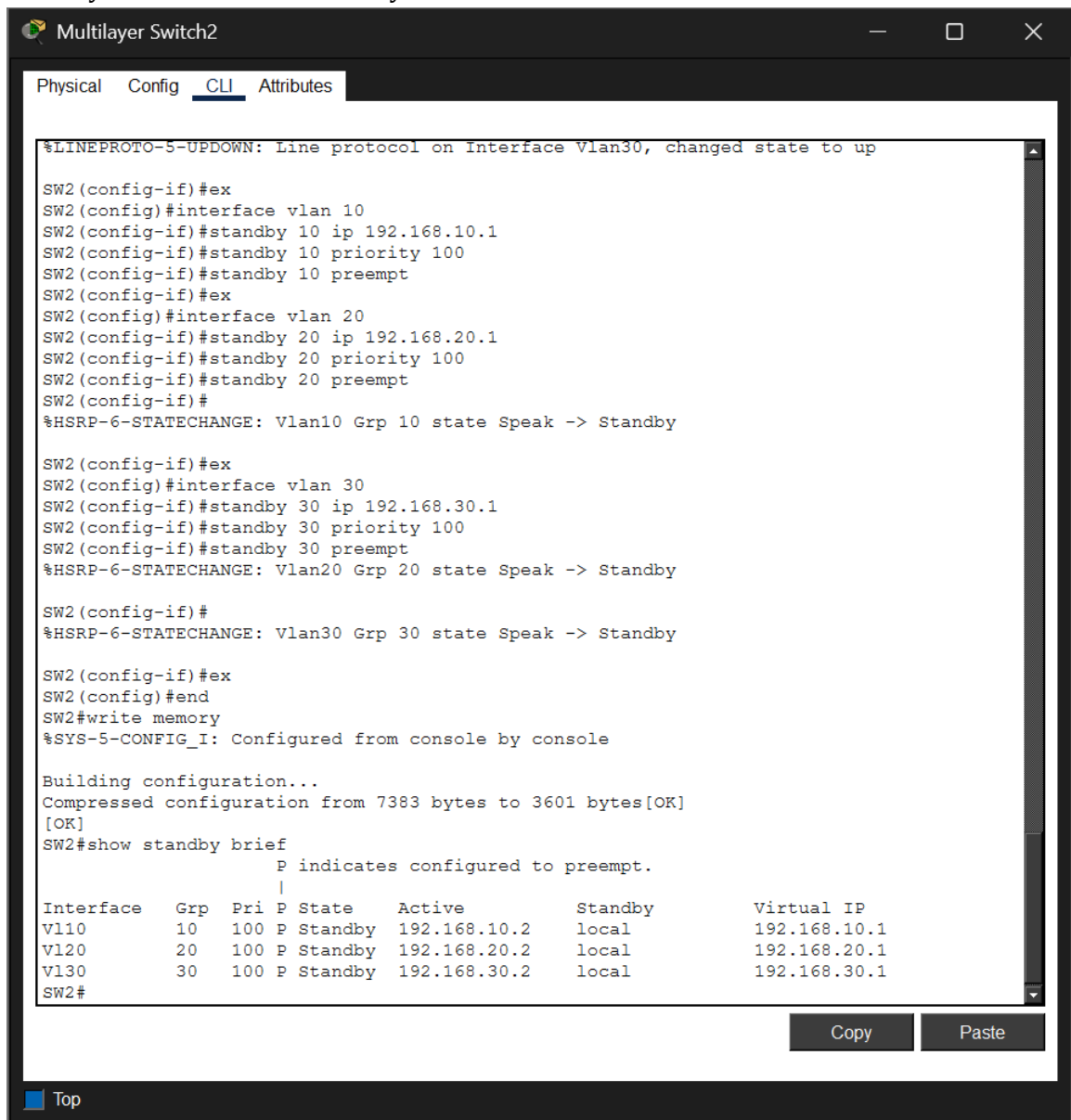
Building configuration...
[OK]
SW-Access#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	Admin	active	Fa0/1
20	Students	active	Fa0/2
30	Servers	active	Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

SW-Access#

CopyPaste

Multilayer switch 2: show standby brief



The screenshot shows a terminal window titled "Multilayer Switch2" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the following commands and output:

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up

SW2(config-if)#ex
SW2(config)#interface vlan 10
SW2(config-if)#standby 10 ip 192.168.10.1
SW2(config-if)#standby 10 priority 100
SW2(config-if)#standby 10 preempt
SW2(config-if)#ex
SW2(config)#interface vlan 20
SW2(config-if)#standby 20 ip 192.168.20.1
SW2(config-if)#standby 20 priority 100
SW2(config-if)#standby 20 preempt
SW2(config-if)#
%HSRP-6-STATECHANGE: Vlan10 Grp 10 state Speak -> Standby

SW2(config-if)#ex
SW2(config)#interface vlan 30
SW2(config-if)#standby 30 ip 192.168.30.1
SW2(config-if)#standby 30 priority 100
SW2(config-if)#standby 30 preempt
%HSRP-6-STATECHANGE: Vlan20 Grp 20 state Speak -> Standby

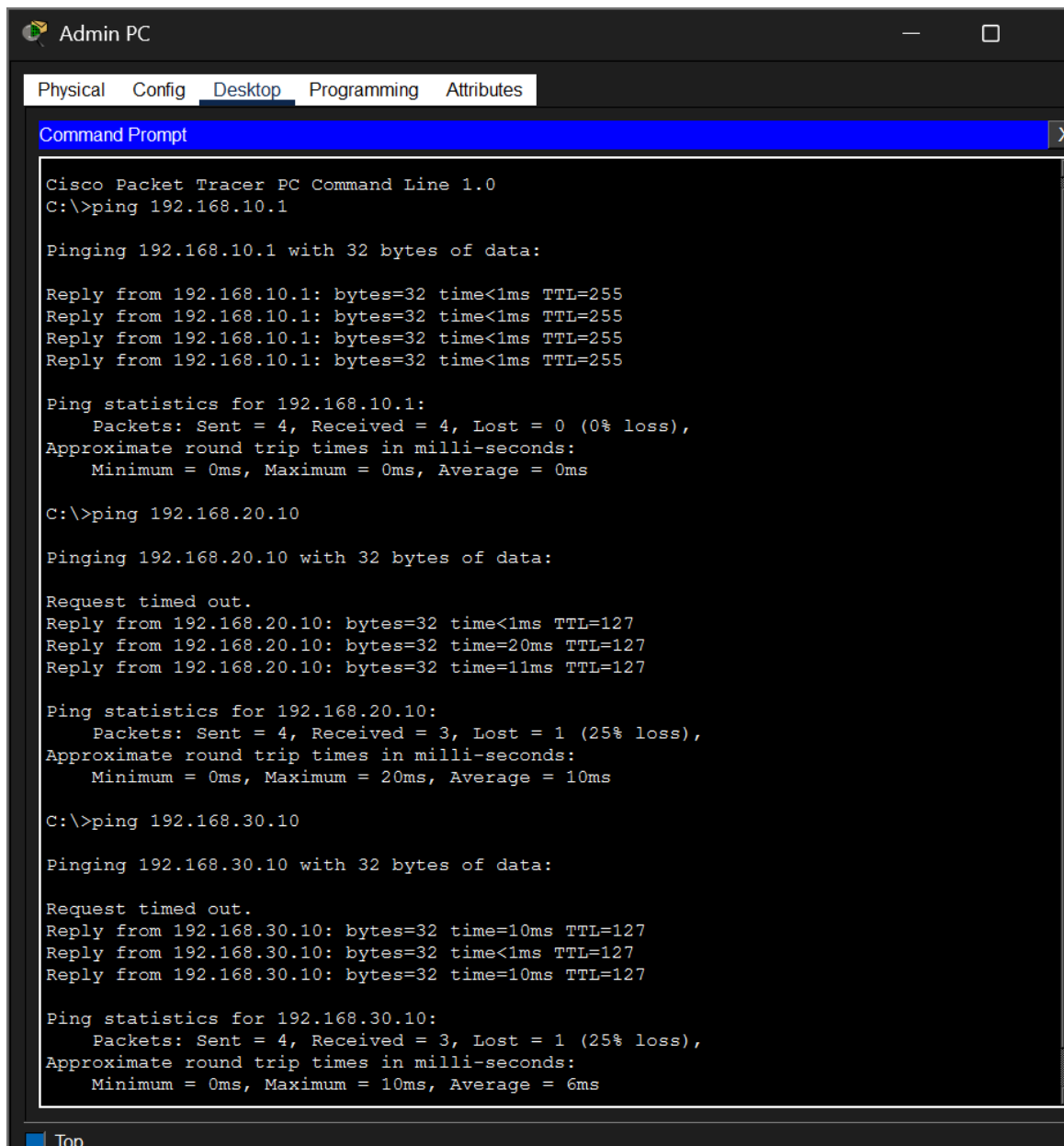
SW2(config-if)#
%HSRP-6-STATECHANGE: Vlan30 Grp 30 state Speak -> Standby

SW2(config-if)#ex
SW2(config)#end
SW2#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
SW2#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri  P State      Active        Standby        Virtual IP
Vl10       10   100  P Standby    192.168.10.2   local          192.168.10.1
Vl20       20   100  P Standby    192.168.20.2   local          192.168.20.1
Vl30       30   100  P Standby    192.168.30.2   local          192.168.30.1
SW2#
```

At the bottom of the window, there are "Copy" and "Paste" buttons, and a "Top" link.

- Ping test results



The screenshot shows a Cisco Packet Tracer PC Command Line window for an 'Admin PC'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The command prompt shows the following sequence of commands and outputs:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time=20ms TTL=127
Reply from 192.168.20.10: bytes=32 time=11ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 10ms

C:\>ping 192.168.30.10

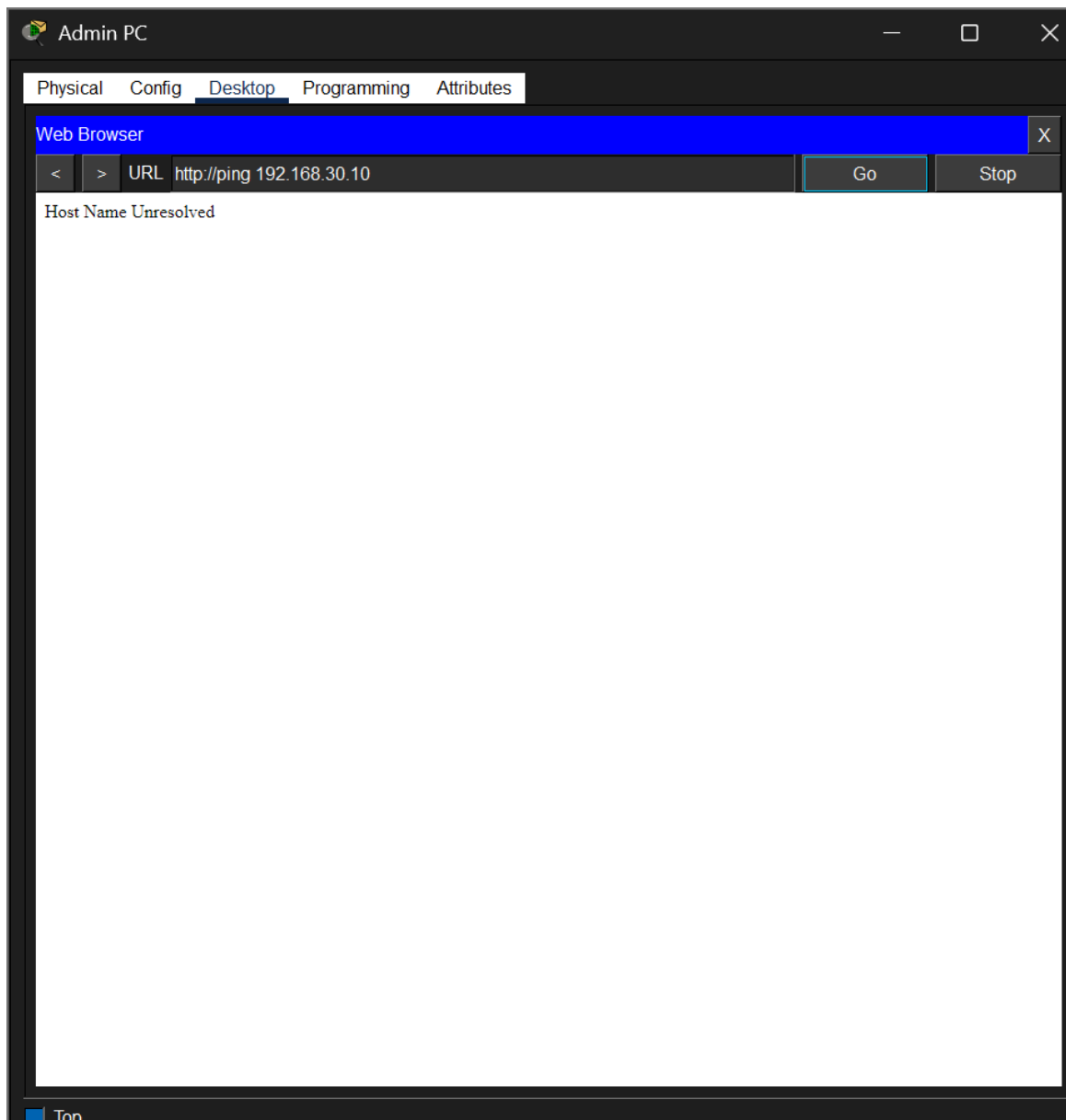
Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time=10ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=10ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 6ms
```

At the bottom of the window, there is a 'Top' button.

- HTTP test proof



- Brief explanation of HSRP/VRRP

HSRP (Hot Standby Router Protocol (Cisco specific)) and VRRP (Virtual Router Redundancy Protocol) are network redundancy protocols used to ensure continuous network connectivity at the default gateway level. A virtual IP address is created and used as the default gateway for hosts. One device is elected as the Active (HSRP) / Master (VRRP) router that forwards traffic. A second device acts as the Standby (HSRP) / Backup (VRRP) router. If the active device fails, the standby device automatically takes over the virtual IP and continues routing traffic.

Week 8 Lab – Networking Device Management (10 Marks)

Objective

Learn how to manage and recover Cisco networking devices by performing:

- Factory reset
- Password recovery
- Configuration backup
- IOS backup and recovery
- Evidence collection (screenshots and config files)

Recommended Environment

You can perform this lab using:

- Cisco Packet Tracer
- GNS3
- Cisco IOSv routers/switches
- Physical Cisco devices

Part 1 – Initial Device Setup

Step 1: Create Basic Topology

PC ----- Switch ----- Router

Step 2: Configure Basic Connectivity

Router Configuration:

```
enable
configure terminal
hostname R1
interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
enable secret Cisco123
line console 0
password Console123
login
line vty 0 4
password Telnet123
login
end
write memory
```

Switch Configuration:

```
enable
configure terminal
hostname SW1
interface vlan 1
ip address 192.168.1.2 255.255.255.0
no shutdown
ip default-gateway 192.168.1.1
enable secret Cisco123
end
write memory
```

PC Configuration:

IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

Part 2 – Factory Reset

Router Factory Reset Steps:

1. Erase startup configuration:

```
write erase
```

OR

```
erase startup-config
```

2. Reload device:

```
reload
```

3. Skip initial configuration dialog by typing:

```
no
```

4. Verify configuration removal:

```
show running-config
```

Part 3 – Password Recovery

Router Password Recovery Procedure:

1. Reload router:

```
reload
```

2. Interrupt boot process and enter ROMMON mode.

3. Ignore startup configuration:

```
confreg 0x2142
```

```
reset
```

4. Enter privileged EXEC mode:

enable

5. Copy startup config to running config:

copy startup-config running-config

6. Set new passwords:

configure terminal

enable secret NewPassword123

line console 0

password ConsoleNew123

login

end

7. Restore normal boot setting:

configure terminal

config-register 0x2102

end

8. Save configuration:

write memory

reload

Part 4 – Configuration Backup

Using TFTP Server:

1. Configure TFTP server with IP 192.168.1.100

2. Verify connectivity:

ping 192.168.1.100

3. Backup running configuration:

copy running-config tftp

4. Backup startup configuration:

copy startup-config tftp

Part 5 – IOS Image Backup

1. Check IOS file:

show flash

2. Backup IOS to TFTP:

copy flash tftp

Part 6 – IOS Recovery

1. Enter ROMMON mode.

2. Configure network variables:

IP_ADDRESS=192.168.1.1

IP_SUBNET_MASK=255.255.255.0

DEFAULT_GATEWAY=192.168.1.254

TFTP_SERVER=192.168.1.100

TFTP_FILE=c1900-universalk9.bin

3. Start recovery:

tftpdnld

4. Reload device:

reset

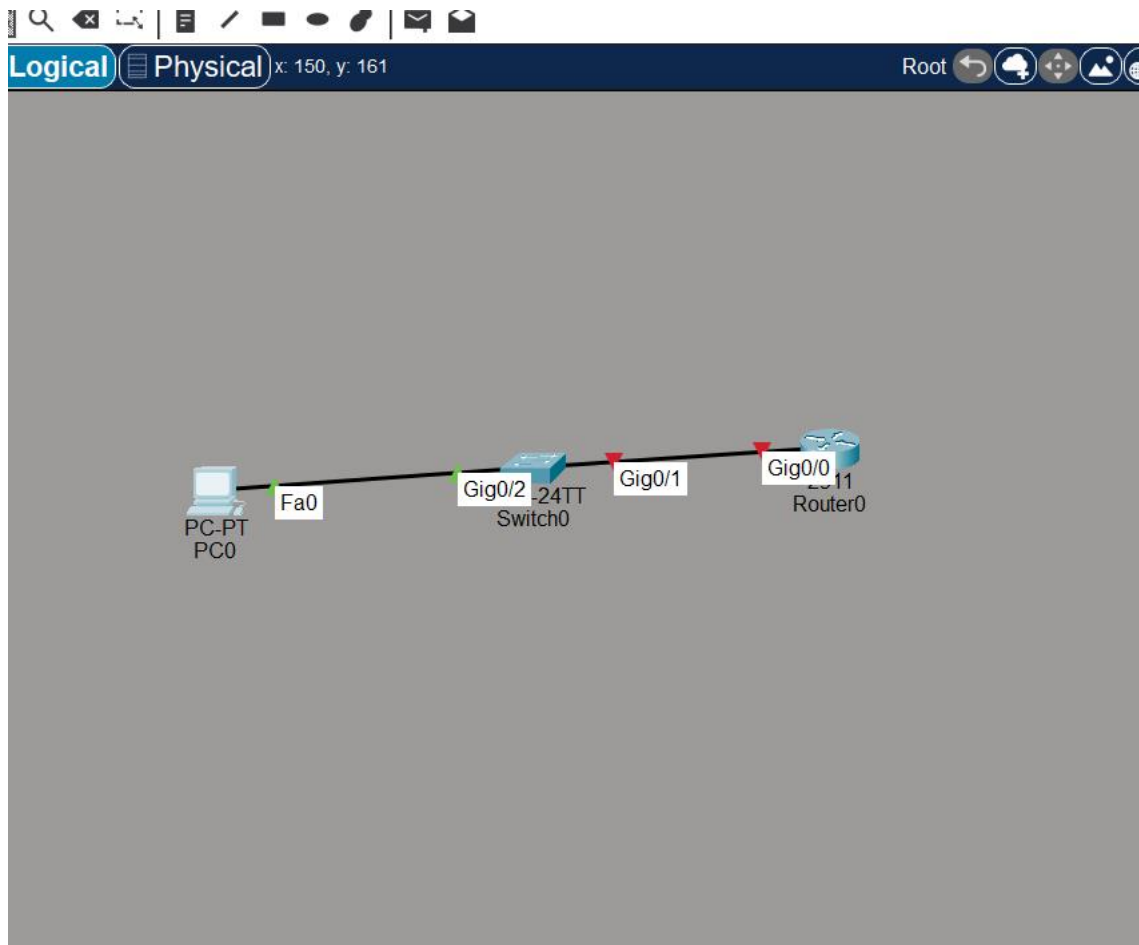
Verification Commands

- show running-config
- show startup-config
- show version
- show flash
- show ip interface brief

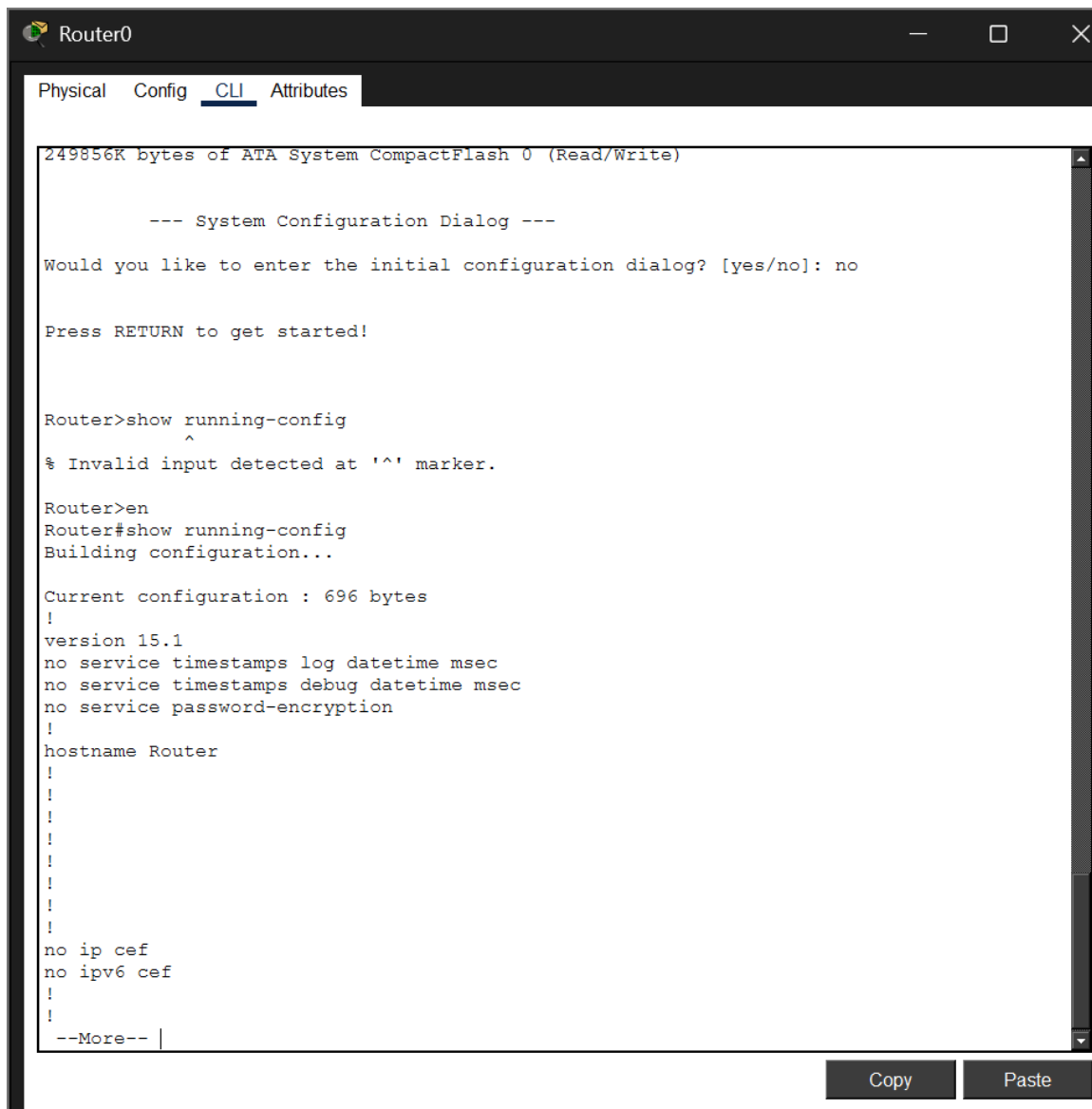
Evidence to Submit

Required Screenshots:

- Initial topology



- Factory reset process



The screenshot shows a terminal window titled "Router0" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active. The terminal output shows the following sequence of commands and responses:

```
249856K bytes of ATA System CompactFlash 0 (Read/Write)

    --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>show running-config
      ^
% Invalid input detected at '^' marker.

Router>en
Router#show running-config
Building configuration...

Current configuration : 696 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
--More-- |
```

At the bottom right of the window, there are "Copy" and "Paste" buttons.

- Password recovery steps

```
Router>en
Router#copy startup-config running-config
%% Non-volatile configuration memory invalid or not present
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#enable secret NewPassword123
Router(config)#line console 0
Router(config-line)#password ConsoleNew123
Router(config-line)#login
Router(config-line)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO2911/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 72/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test

Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x3bcd3d8
Self decompressing the image :
##### [OK]
Smart Init is enabled
smart init is sizing iomem

          TYPE          MEMORY_REQ
Onboard devices &
  buffer pools          0x022F6000
```

Copy Paste

- Successful ping to TFTP server


```
Router0
Physical Config CLI Attributes

Vlan1 unassigned YES NVRAM administratively down down
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#ex
Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 192.168.1.1 YES manual up up
GigabitEthernet0/1 unassigned YES NVRAM administratively down down
GigabitEthernet0/2 unassigned YES NVRAM administratively down down
Vlan1 unassigned YES NVRAM administratively down down
Router#ping 192.168.1.100

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.100, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/6/12 ms

Router#copy running-config tftp
Address or name of remote host []? 192.168.1.100
Destination filename [Router-config]?

Writing running-config...!!
[OK - 709 bytes]

709 bytes copied in 0.013 secs (54538 bytes/sec)
Router#copy startup-config tftp
Address or name of remote host []? 192.168.1.100
Destination filename [Router-config]?

Writing startup-config...!!
[OK - 776 bytes]

776 bytes copied in 0 secs
Router#
```

- Configuration backup success

Router0

Physical Config CLI Attributes

```
rommon 5 > TFTP_FILE=c1900-universalk9-mz.SPA155-3.M4a.bin
rommon 6 > tftpdnld

      IP_ADDRESS: 192.168.1.1
      IP_SUBNET_MASK: 255.255.255.0
      DEFAULT_GATEWAY: 192.168.1.254
      TFTP_SERVER: 192.168.1.100
      TFTP_FILE: c1900-universalk9-mz.SPA155-3.M4a.bin
Invoke this command for disaster recovery only.
WARNING: all existing data in all partitions on flash will be lost!

Do you wish to continue? y/n: [n]: y

.....[TIMED OUT]
TFTP: Operation terminated.

rommon 7 > FE_SPEED_MODE=4
rommon 8 > tftpdnld

      IP_ADDRESS: 192.168.1.1
      IP_SUBNET_MASK: 255.255.255.0
      DEFAULT_GATEWAY: 192.168.1.254
      TFTP_SERVER: 192.168.1.100
      TFTP_FILE: c1900-universalk9-mz.SPA155-3.M4a.bin
Invoke this command for disaster recovery only.
WARNING: all existing data in all partitions on flash will be lost!

Do you wish to continue? y/n: [n]: y

.....[TIMED OUT]
TFTP: Operation terminated.

rommon 9 > System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO2911/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 72/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized
```

Copy Paste

Top

- show version output

```
Router0
Physical Config CLI Attributes

Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ex
Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router# show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thurs 5-Jan-12 15:41 by pt_team

ROM: System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
cisco2911 uptime is 10 minutes, 58 seconds
System returned to ROM by power-on
System image file is "flash0:c2900-universalk9-mz.SPA.151-1.M4.bin"
Last reload type: Normal Reload

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
--More--

Copy Paste
Top
```

- show flash output

Router0

Physical Config CLI Attributes

DRAM configuration is 64 bits wide with parity disabled.
 255K bytes of non-volatile configuration memory.
 249856K bytes of ATA System CompactFlash 0 (Read/Write)

License Info:

License UDI:

Device#	PID	SN
*0	CISCO2911/K9	FTX15246BTY-

Technology Package License Information for Module:'c2900'

Technology	Technology-package Current	Type	Technology-package Next reboot
ipbase	ipbasek9	Permanent	ipbasek9
security	disable	None	None
uc	disable	None	None
data	disable	None	None

Configuration register is 0x2142

Router#

Router#show flash

System flash directory:

File	Length	Name/status
3	33591768	c2900-universalk9-mz.SPA.151-4.M4.bin
2	28282	sigdef-category.xml
1	227537	sigdef-default.xml

[33847587 bytes used, 221896413 available, 255744000 total]

249856K bytes of processor board System flash (Read/Write)

Router#

Copy Pa

Required Files:

- Running configuration file

Router0

Physical Config CLI Attributes

data	disable	None	None
------	---------	------	------

Configuration register is 0x2142

Router#
Router#show flash

System flash directory:

File	Length	Name/status
3	33591768	c2900-universalk9-mz.SPA.151-4.M4.bin
2	28282	sigdef-category.xml
1	227537	sigdef-default.xml

[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)

Router#show running-config
Building configuration...

Current configuration : 722 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
--More--

Copy Paste

- Startup configuration file

Router0

Physical Config CLI Attributes

```
!
no ip cef
no ipv6 cef
!
!

Router#show flash

System flash directory:
File      Length      Name/status
  3    33591768  c2900-universalk9-mz.SPA.151-4.M4.bin
  2     28282    sigdef-category.xml
  1    227537    sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)

Router#show startup-config
Using 776 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
enable secret 5 $1$mERr$1oJ053xwDS1EWhg2PBRFF0
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
--More--
```

Copy Paste

- IOS backup file (if possible)

Router0

Physical Config CLI Attributes

```
1 227537 sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)

Router#show running-config
Building configuration...

Current configuration : 722 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!

Router#show flash

System flash directory:
File Length Name/status
 3 33591768 c2900-universalk9-mz.SPA.151-4.M4.bin
 2 28282 sigdef-category.xml
 1 227537 sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)

Router#
```

Copy Paste

Concepts Being Tested

- Factory Reset – Restores device defaults
- Password Recovery – Regains administrator access
- TFTP Backup – Remote storage of configurations

- IOS Recovery – Restores corrupted IOS images
- Flash Memory – Stores IOS images
- ROMMON – Low-level recovery mode

Expected Learning Outcomes

After completing this lab, students should be able to:

- Recover forgotten passwords
- Reset Cisco devices safely
- Back up and restore configurations
- Manage IOS images
- Perform disaster recovery procedures
- Use TFTP for device management